Basic concepts

Throughout we will assume that M is a C^∞-manifold, very often M will also be compact.

Def A symplectic structure on a mfld M is a non-degenerate closed 2-form $\omega \in \Omega^2(M)$

Recall that non-degeneracy means that $\forall q \in M$, $\forall v \neq 0 \in T_qM$, $\exists y \in T_qM$ s.t. $\omega_q(v,y) \neq 0$. This implies that each tangent space (T_qM, ω_q) is a symplectic vector space. The manifold M is necessarily even-dimensional and we have the following:

Prop Let (M^{2n}, ω) be symplectic, then $\omega^n := \omega \wedge \dots \wedge \omega \in \Omega^{2n}(M)$ never vanishes, hence M is orientable

Proof It's a linear algebra fact: if V is a $2n$-dim linear real v-s, then a skew-symmetric form ω on V is non-degenerate iff the n-fold exterior power is non-zero. In fact assume that ω is degenerate, let $v \neq 0$ s.t. $\omega(v,y) = 0 \ \forall y$. Now choose a basis $v, v_2, \dots, v_{2n}$ of V. Then we claim that $\omega^n(v, v_2, \dots, v_{2n}) = 0$. By induction on n in fact, we have that if $n=1$, then $\omega(v, v_2) = 0$ by assumptions on v. Now assume $\omega^{n-1}(v, v_2, \dots, v_{2n-2}) = 0$, then

$$\omega \wedge \omega^{n-1}(v, v_2, \dots, v_{2n-2}, v_{2n-1}, v_{2n}) = \lambda \cdot \mathrm{Alt}\left(\omega \otimes \omega^{n-1}\right)$$
$$= \lambda' \cdot \sum_{\sigma \in S_{2n}} \mathrm{sgn}(\sigma)\, \omega \otimes \omega^{n-1}(v_{\sigma_1}, \dots, v_{\sigma(2n)})$$

It's easy to see that the last summand is always 0, since v is either in the first 2 slots (hence killing ω) or is in the last $2n-2$ (and killing ω^{n-1} by inductive hp.)

Conversely, suppose that ω is non-degenerate, then by picking a symplectic basis one can build $\Phi: V \xrightarrow{\cong} V$ s.t. $\Phi^*\omega = \omega_{std}$, and it's immediate to see that ω_0^n is a volume form. $\square$

Is the symplectic property an exclusively "cohomological" property?

This question was only recently answered, more precisely it was not known whether any compact even-dim manifold M which admits a non-degenerate 2-form $\tau \in \Omega^2(M)$ and a suitable cohomology $a \in H^2(M;\mathbb{R})$, admits a symplectic structure which represents the class a and is isotopic, through non-degenerate forms, to the given form τ. By suitable we mean that $a^n > 0$ in $H^{2n}(X;\mathbb{R}) \cong \mathbb{R}$ (Notice that this is far weaker than $\tau^{2n} \neq 0 \ \forall (v_1, \dots, v_{2n}) \neq 0$)

Methods due to Donaldson and Taubes now permit the construction of explicit manifolds that admit a suitable pair (z,a) but which do <u>not</u> have any symplectic structure.

For open manifolds (i.e. connected manifolds w/o boundary and no compact component) these questions are completely answered by Gromov's h-principle. His theorem asserts that for every almost complex open mfld M and every cohomology class $a \in H^2(M;\mathbb{R})$, the inclusion of the space of symplectic forms representing the class a into the space of non-degenerate forms on M is an homotopy equivalence.

Another classical definition is the following

Def A symplectomorphism between two symplectic manifolds (M_1,ω_1) and (M_2,ω_2) is a diffeomorphism $\varphi: M_1 \to M_2$ s.t.

$$\omega_2 = \varphi^* \omega_1$$

Examples the first example of a symplectic manifold is $\mathbb{R}^{2n}$ itself with the standard symplectic form.

Orientable closed surfaces are symplectic.

Using the previous proposition we immediately have that S^n for $n>2$ is not symplectic, since $H^2(S^n;\mathbb{R})=0$ and a volume form on a closed mfld induces a fundamental class in homology. Similarly we see that $\mathbb{CP}^n$ is a symplectic manifold (actually more is true, $\mathbb{CP}^n$ is Kähler).

One of the most important example of symplectic manifold is the cotangent bundle of any manifold. These are a fundamental case to understand since they appear very often as the "local" form of our symplectic manifold.

A cotangent bundle T^*L is the vector bundle whose sections are 1-forms on L and so it carries a universal 1-form $\lambda_{can} \in \Omega^1(T^*L)$ characterized as follows:

Prop the 1-form $\lambda_{can} \in \Omega^1(T^*L)$ is uniquely characterized by the property that

$$\sigma^* \lambda_{can} = \sigma \quad \forall \text{ 1-form (seen as a section } \sigma: L \to T^*L)$$

It's not hard to convince ourselves that in standard coordinates (x,y), where $x \in \mathbb{R}^n \subset L$ and

$y \in \mathbb{R}^n \subset T_xL$, we have $\lambda_{can} = y\,dx$.

Now we define $\omega_{can} = -d\lambda_{can}$ (the minus is there for some sign convention).

Prop (TL, ω_{can}) is a symplectic manifold

Proof It's enough to work locally, hence $\omega_{can} = -d\lambda_{can}$

$$= dx \wedge dy$$

and it's immediate to see that it's non-degenerate and closed. $\square$

Darboux Theorem

We will begin by describing Moser's argument on the isotopy of symplectic forms. This has many important consequences: symplectic structures have no local invariants. The word local here has many possible interpretations: one can localize at a point and get Darboux's theorem, one can localize near a symplectic or Lagrangian submanifold, or one can localize w.r.t. time and consider properties of smooth deformations of a symplectic structure.

Roughly speaking, Moser's argument shows that for every family of symplectic forms $\omega_t \in \Omega^2(M)$ with an exact derivative
$$\frac{d}{dt}\omega_t = d\sigma_t$$
there exists a family of diffeomorphisms $\psi_t \in \mathrm{Diff}(M)$ s.t.
$$\psi_t^*\omega_t = \omega_0 \qquad (1)$$
The key idea is to determine the diffeomorphism ψ_t by representing them as the flow of a family of vector fields X_t on M.

Thus we suppose that
$$\frac{d}{dt}\psi_t = X_t \circ \psi_t \qquad \psi_0 = Id \qquad (2)$$
The vector fields X_t have to be constructed so that (1) is satisfied.

Let us differentiate (1) w.r.t. time

$$0 = \frac{d}{dt}\psi_t^*\omega_t \overset{\text{chain rule}}{=} \underbrace{\psi_t^*\frac{d}{dt}\omega_t} + \underbrace{\frac{d}{dt}\psi_t^*\omega_t} \qquad \text{Cartan magic formula + naturality}$$

$$= \underbrace{\psi_t^*\frac{d}{dt}\omega_t} + \underbrace{\psi_t^*\iota_{X_t}d\omega_t + \psi_t^* d\iota_{X_t}\omega_t}$$

Since ω_t is closed $\forall t$ and its derivative is exact, this identity will be satisfied if

$$0 = \varphi_t^* d\sigma_t + \varphi_t^* d\iota_{X_t}\omega_t$$
$$= \varphi_t^* d(\underbrace{\sigma_t + \iota_{X_t}\omega_t}_{(*)})$$

We put $(*)$ equal to zero and get

$$0 = \sigma_t + \iota_{X_t}\omega_t \qquad (3)$$

Since ω_t is non-degenerate there exists a unique family of vector fields X_t satisfying (3). Hence the diffeomorphisms φ_t determined by (2) satisfy (1) as required.

This argument is perfectly fine when M is compact, in general one has to make sure that the solutions of the diff. equations (2) exist on the required time interval.

Lemma 1 Let M be a $2n$-dimensional smooth mfld and $Q \subset M$ be a compact submanifold. Suppose that ω_0 and $\omega_1 \in \Omega^2(M)$ are closed 2-forms s.t. at each point q of Q the forms ω_0 and ω_1 are equal and non-degenerate on T_qM.

Then there exist an open nbhd N_0 and N_1 of Q and a diffeomorphism $\varphi: N_0 \to N_1$ s.t.

$$\varphi_{|Q} = Id \qquad \varphi^*\omega_1 = \omega_0$$

Proof We will give only a sketch. The proof itself is very straightforward but not that enlightening.

Assume we prove that $\exists$ a 1-form $\sigma \in \Omega^1(N_0)$ s.t.

$$\sigma_{|T_qM} = 0 \qquad d\sigma = \omega_1 - \omega_0$$

Then we can consider the family of closed forms on N_0

$$\omega_t = \omega_0 + t(\omega_1 - \omega_0) = \omega_0 + t\,d\sigma$$

Shrinking N_0 if necessary we may assume that ω_t is non degenerate in N_0 $\forall t$. Thus we can solve eq (3) and the resulting vector fields X_t vanish on Q. WLOG in N_0 the solutions of (1) exist on the required interval $0 \le t \le 1$. By applying Moser's argument we conclude. $\square$

Corollary [Darboux theorem] Every symplectic form ω on M is locally diffeomorphic to the standard form ω_0 on $\mathbb{R}^{2n}$

Proof Apply lemma 1 to $Q = \{pt.\}$ □

Another useful consequence of lemma 1 is the following.

Corollary For $j = 0,1$ let (M_j, ω_j) be a symplectic manifold with compact symplectic submanifold Q_j. Suppose that there is an isomorphism $\Phi : \nu_{Q_0} \to \nu_{Q_1}$ of the symplectic normal bundles to Q_0 and Q_1 which covers a symplectomorphism $\phi : (Q_0, \omega_0) \to (Q_1, \omega_1)$. Then ϕ extends to a symplectomorphism $\psi : (\mathcal{N}(Q_0), \omega_0) \to (\mathcal{N}(Q_1), \omega_1)$ such that $d\psi$ induces the map Φ on $\nu_{Q_0} = (TQ_0)^{\omega}$.

Corollary [Lagrangian neighbourhood theorem] Let (M, ω) be a symplectic manifold and $L \subset M$ be a compact Lagrangian submanifold. Then there exists a nbhd $\mathcal{N}(L_0) \subset T^*L$ of the zero section, a nbhd $V \subset M$ of L, and a diffeomorphism $\phi : \mathcal{N}(L_0) \to V$ s.t.

$$\phi^* \omega = -d\lambda_{can} \qquad \phi|_L = id$$

where λ is the canonical 1-form on T^*L □